ADNOC GAS OPERATIONS & MARKETING LLC

PRESERVATION PLAN

EPCm CONTRACT NO.: 3000076501
ADNOC GAS PROJECT NO.: 30201-50150-LT-HE

LONG TERM - EXECUTE STAGE
PROJECT 5.0 (P5)
EPCm FOR HABSHAN DBN

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1	17	45	Preservation Sequencing

1.0 INTRODUCTION

1.1 BACKGROUND

ADNOC aims to increase oil production to produce 5.0 MMBOPD by year-end 2027. Accordingly, associated gas production is anticipated to increase. Also, to meet the future oil and gas demand, non-associated gas developments are considered. Hence, the overall gas supply scenarios for major ADNOC Gas (hereafter referred to as COMPANY) assets i.e., Das Facilities, Bab and Habshan Complex (Habshan 0, 1, 2, 4, and 5), Asab and Bu Hasa, is anticipated to change depending on their feed sources, profiles including rates, compositions, and operating conditions. These changes will also subsequently impact the COMPANY downstream facilities, particularly, products handling and transferring facilities i.e., Sales Gas Network, NGL Pipeline Network, Ruwais Fractionation Trains (1, 2, 3, 4 and upcoming new Train 5 part of Bab Cap Gas SELECT Study), Sulphur Handling Facilities, and Condensate systems, etc.

COMPANY has carried out the Long Term DEFINE for the project with M/s Amec International Ltd. (M/s Wood) and subsequently decided to proceed with Long Term EXECUTE phase execution to ensure the readiness of COMPANY facilities to accommodate increased gas supply by December 2027.

Accordingly, COMPANY has awarded EPCm contract to M/s Amec International Ltd. (M/s Wood) to carry out the Long Term scope requirements. M/s Wood shall act as a single entity in the performance of the services and a single point of accountability to ADNOC Gas for the successful completion of Long Term EXECUTE phase.

1.2 LONG TERM EPCM PROJECT OBJECTIVES

The main objective of the Long Term EPCm is to execute the Habshan DBN Project to support ADNOC to achieve oil production target of 5.0 million BOPD so that Associated and Non-Associated Gas can be processed within the overall processing capacity of Habshan. The Long Term EPCm scope includes the following major areas as a minimum:

1. Habshan Complex (H0, H1 Train 1/2, H2, H4, H3 for SS-300 Tie-in)
2. Bab CDS and Bab NGL
3. Habshan 5
4. Pipeline scope activities including the new pipelines below
   I. Bab CDS to Habshan Complex HP pipeline
   II. Bab CDS to Habshan Complex MP pipelines (2 no.s)
   III. Habshan 5 to Habshan 2 pipeline

The scope also includes adequacy checks for Ruwais NGL Fractionation Trains, MERAM,

Habshan Sulphur Granulation Plant, Sales gas, NGL, Condensate and interconnecting Pipeline network and Utilities for all impacted AG facilities.

The Long Term EPCM scope includes the following major activities as a minimum:

1. Detailed Engineering and design services
2. Procurement Services
3. Construction Management services
4. Pre-commissioning, Commissioning, Start-up services
5. Final documentation handover and Project close-out

Project also involves major interface activities with other on-going projects. Refer Project Interface management plan (Doc Num: PP-30201_50150_LT_HE-GG-000-50-00-006) for a complete list of interface projects.

1.3 SCOPE BATTERY LIMIT

Refer Design Basis (Doc Num: BOD-30201_50150_LT_HE-GG-000-00-00-001) for the scope battery limits, Gas allocation and detailed Design basis.

2.0 PURPOSE OF THIS DOCUMENT

This document provides the requirements to ensure that all materials received and stored at the MODULE FABRICATORS and at the HABSHAN Facility is in the appropriate condition at the time of start-up and operations.

This procedure defines the minimum requirements for material and equipment preservation in the temporary lay down yards prior to assembling / installation and during installation at either the module fabrication yard or the HABSHAN complex construction site.

Special storage requirements as detailed in the SUPPLIER's data books and instruction manuals shall be adhered to. Instruction indicated on original purchase order, SUPPLIER's specification, etc. shall be followed. In case of inconsistency with project specification, the more stringent requirement shall be applied.

The appropriate condition means:

- ✓ The equipment and materials meet, in full, the specifications, datasheets, and all requirements of the purchase order.
- ✓ The equipment and materials have not been allowed to deteriorate at any time during the preservation period.
- ✓ The appropriate condition also applies to the appearance, cleanliness, and safety of the equipment and materials. The state is determined at any inspection, scheduled or unscheduled.

- ✓ Use of the assets means the first delivery of gas unless specified otherwise. Therefore, the requirements apply from the manufacturing facility, through the construction, commissioning, and any period until start-up.

Whereas this document specifies preservation up to various points in the development life cycle, the actual handover of responsibility for preservation from and to multiple entities will be determined by the Preservation Team and specified in the relevant contracts and purchase orders. This document does not define these handover points.

3.0 ORDER OF DOCUMENT PRECEDENCE

In the event of any conflict between the applicable documents, the order of precedence shall be as follows: the ANNEXURES or among the ANNEXURES, the order of precedence below shall prevail and shall guide the construction of any part of the AGREEMENT:

1. (a) this FORM OF AGREEMENT;
2. (b) ANNEXURE 1 – SPECIAL CONDITIONS;
3. (c) ANNEXURE 2 – GENERAL TERMS AND CONDITIONS;
4. (d) ANNEXURE 11 – COMPLETION AND ACCEPTANCE REQUIREMENTS
5. (e) ANNEXURE 15 – RESPONSIBILITIES MATRIX;
6. (f) ANNEXURE 13 – ICV IMPROVEMENT PLAN;
7. (g) ANNEXURE 12 – HSE REQUIREMENTS;
8. (h) ANNEXURE 3 – SCOPE OF WORK;
9. (i) ANNEXURE 7 – INSURANCE;
10. (j) ANNEXURE 10 - PROJECT EXECUTION REQUIREMENTS; k) ANNEXURE 14 - QUALITY CONTROL AND INTEGRITY MANAGEMENT (QA/QC);
11. (l) ANNEXURE 16 - KPI SCHEDULE;
12. (m) ANNEXURE 5 - PRICING SCHEDULE;
13. (n) ANNEXURE 17 - TOTAL INSTALLED COST;
14. (o) ANNEXURE 4 - COMPLETION SCHEDULE;
15. (p) ANNEXURE 9 - CONTRACTOR' S SUBMISSIONS.
16. (q) ANNEXURE 6 - COMPANY PROVIDED FACILITIES;

17. (r) ANNEXURE 8 – SECURITY AND FORMS;
18. (s) ANNEXURE 18 – FORM OF EPCM SUPPLY AGREEMENT; and
19. (t) ANNEXURE 19 – FORM OF EPCM CONSTRUCTION CONTRACT.

4.0 REFERENCE, CODES AND STANDARDS

4.1 REFERENCE DOCUMENTS – Preservation ITRs listing

Table 4-1: Reference Documents

Name	Description	Discipline	APPLICABLE	Discipline
A05P	Building, Accommodation Unit	A	Yes	Architectural
B01P	Plant Room, LER, Switch Room	B	Yes	Buildings
E01P	Electrical Panels and Equipment	E	Yes	Electrical
E02P	Electrical Transformer	E	Yes	Electrical
E03P	Electrical Batteries, Chargers, UPS	E	Yes	Electrical
E04P	Electrical Motors and Generators	E	Yes	Electrical
E05P	Insulation Resistance Test	E	Yes	Electrical
E06P	AntiCon Space Heater Test	E	Yes	Electrical
E07P	LV MV Switchgear	E	Yes	Electrical
E08P	LV HV Bus Duct	E	Yes	Electrical
E09P	L&SP, Small Power Devices, RCUs	E	Yes	Electrical
E10P	LV MCC	E	Yes	Electrical
G02P	Equipment Skids and Packages	G	Yes	Safety
H01P	Chillers, AHUs, Dampers	H	Yes	HVAC
H02P	HVAC Fans	H	Yes	HVAC
H03P	HVAC Systems	H	Review	HVAC
I02P	Actuated Control Valves	I	Review	Instrumentation
I03P	HPUs, PPUs, Control Panels	I	Yes	Instrumentation
I05P	Field Instruments	I	Review	Instrumentation
L02P	Hose Reels, Hydrants, Foam Units, Extinguishers	L	Yes	Lifeboat
L03P	Safety Showers and Cabinets	L	Yes	Lifeboat
L04P	Deluge and Sprinkler Valve Sets	L	Yes	Lifeboat
M01P	Rotating Equipment - General Pumps and Gearboxes	M	Yes	Mechanical
M02P	Rotating Equipment - Fans and Blowers	M	Yes	Mechanical
M03P	Rotating Equipment - Compressors	M	Yes	Mechanical
M04P	Rotating Equipment - Gas Turbines	M	Yes	Mechanical
M05P	Rotating Equipment - Diesel Engines	M	Yes	Mechanical
M06P	Equipment Shaft rotations	M	Yes	Mechanical
M07P	Static Equipment - Tanks, Vessels, Exchangers	M	Review	Mechanical
M08P	Lifting Equipment - General	M	Review	Mechanical

M10P	Lifting Equipment - Overhead Cranes	M	Yes	Mechanical
M11P	Fin Fan Air Coolers	M	Yes	Mechanical
M12P	Lube Oil Skid	M	Yes	Mechanical
T02P	CCTV	T	Yes	Telecomms

Sl. No.	Document Number	Document Title
1.	PP-30201_50150_LT_HE-GG-000-53-00-002	Pre-commissioning Plan
2.	PP-30201_50150_LT_HE-GG-000-53-00-003	PRE-COMMISSIONING RESPONSIBILITY MATRIX
3.	PP-30201_50150_LT_HE-GG-000-53-00-004	SYSTEMIZATION PHILOSOPHY AND PROCEDURE
4.	RPT-30201_50150_LT_HE-GG-000-53-00-001	MECHANICAL HANDLING STUDY/ RIGGING STUDIES
5.	NM-30201_50150_LT_HE-GG-000-53-00-007	COMMISSIONING LUBRICATION SCHEDULE
6.	PP-30201_50150_LT_HE-GG-000-53-00-012	SPECIALISED FLUSHING AND CLEANING PLAN
7.	PP-30201_50150_LT_HE-GG-000-53-00-015	PIPELINE DRY-OUT PROCEDURE
8.	PP-30201_50150_LT_HE-GG-000-53-00-020	P2A , PROJECT TO ASSET TRANSFER PLAN

4.2 ADNOC SPECIFICATION

ADNOC Specification to be referred as applicable to Project and agreed in Basis For Design Document

4.3 INTERNATIONAL CODES AND STANDARDS

Table 4-2: International Codes and Standards

Sl. No.	Document Number	Document Title
9.	AISC 325	STEEL CONSTRUCTION MANUAL
10.	AISC 326	DETAILING FOR STEEL CONSTRUCTION
11.	AISC 341	SEISMIC PROVISIONS FOR STRUCTURAL STEEL BUILDINGS
12.	AISC 360	SPECIFICATION FOR STRUCTURAL STEEL BUILDINGS
13.	ASCE 7-16 & 7-05	MINIMUM DESIGN LOADS FOR BUILDINGS AND OTHER STRUCTURES
14.	IBC – 2018	INTERNATIONAL BUILDING CODE (IBC) 2018
15.	ADIBC-2013	ABU DHABI INTERNATIONAL BUILDING CODE
16.	ASME B31.3	PROCESS PIPING

5.0 DEFINITIONS

5.1 GENERAL & Preservation DEFINITIONS

For this document, the following general definitions are applicable.

Table 5-1: General Definitions

DEFINITION	DESCRIPTION
COMPANY	ADNOC GAS (AG) for whom the SERVICES are being performed by the EPCm CONTRACTOR.
EPCm CONTRACTOR	Amec International Limited – Abu Dhabi (WOOD) appointed by COMPANY to perform Engineering, Procurement, Construction management, Pre-Commissioning and Commissioning stage for Long Term Solution (under EXECUTE phase).
FACILITIES	Habshan Complex (H0, H1 Train 1/2, H2, H4, H3 for SS-300 Tie-in), Bab CDS, Bab NGL, Habshan 5 and interconnecting pipelines.

DEFINITION	DESCRIPTION
LICENSOR	Supplier of packaged units and responsible for the operating and design performance.
MANUFACTURER/ VENDOR/ SUPPLIER	The Party (parties) which manufactures and/or supplies material, equipment, technical documents or drawings and/or services to perform the duties specified by the EPCm CONSTRUCTION CONTRACTOR.
PMC	Project Management Consultant appointed by COMPANY for the PROJECT
PMT	Project Management Team
PROJECT	PROJECT 5.0 (P5), Long Term EPCm for HABSHAN DBN
SERVICES	The scope of work to be completed during the EXECUTE stage (EPCm) of the PROJECT.
SHALL	The use of the word “shall” indicate a mandatory requirement
SHOULD	The use of the word “should” indicate a strong recommendation to comply with the requirements of this document
EPCm CONSTRUCTION CONTRACTOR	Subcontractor appointed by EPCm CONTRACTOR to carry out Construction works at site for respective construction packages.
SUB-CONTRACTOR	The COMPANY approved party, which carries out services subcontracted by EPCm CONTRACTOR.
Initial Preservation	Initial Preservation is the application of a type of preservation or protection to equipment or materials before delivery to the storage area or construction site. The responsible Supplier or Contractor typically applies it. Initial preservation shall remain in effect until the remote site preservation organization starts the preservation maintenance, or the equipment/material is de-preserved to allow construction, Pre-commissioning, and commissioning to commence.
De-preservation	These are activities that are performed, if required, on equipment or materials to allow Pre-commissioning and commissioning
Re-preservation	These are activities that are performed, if required, after any project activity requiring work on unpreserved equipment or materials or where such action has disrupted prior preservation, re-establishing preservation maintenance

DEFINITION	DESCRIPTION
Procurement Inspection	Delivered materials will be inspected for acceptability at the Site/Warehouse/Laydown Areas by physical inspection of the material and review of the specified documentation. Inspections should check both commercial and technical requirements upon receipt. Field Procurement will perform a commercial inspection of materials for conformance to the purchase order requirements and note any discrepancies on a USOD report.
Procurement Inspection	Delivered materials will be inspected for acceptability at the Site/Warehouse/Laydown Areas by physical inspection of the material and review of the specified documentation. Inspections should check both commercial and technical requirements upon receipt. Field Procurement will perform a commercial inspection of materials for conformance to the purchase order requirements and note any discrepancies on a USOD report.
Preservation Inspection	A Preservation Inspection is issued from Co-Console and represents one maintenance instance for one tagged item. Preservation Inspections are issued via the preservation app that contain a list of preservation and maintenance activities for a tagged item (Attachment).
Preservation Check List	A Preservation Check List is an itemized list defining the preservation requirements for the various types of equipment and materials with preservation periods identified against each specific check.
Preservation Storage Specification (PSS)	These define the specific storage and preservation requirements at the component level.
Preservation Inspection Interval	These define the preservation maintenance intervals per type of equipment (and materials), as identified in the PSS.
Equipment Preservation	Equipment Preservation means the preservation of any type of tagged item delivered to the Project (i.e., individually tagged items or packaged equipment items). The term Tagged Item means listed in the Master Equipment List or Instrument Index or Loop List.

DEFINITION	DESCRIPTION
Bulk Material Preservation	Hereafter, called Material Preservation, means the preservation of untagged items delivered to the project (i.e., bulk deliveries, prefabricated elements, spools, lines, valves, complete systems, etc.). The untagged item also includes manual valves that may have a “type code” or tag.
Storage Specification	All Equipment and Material delivered to the construction sites(s) will have a predetermined storage requirement as indicated on the PSS. Storage requirements reflect the associated preservation requirements. Storage classifications are defined in Table 6.5: Storage Classifications below.

5.2 TECHNICAL DEFINITIONS

For this document, the following technical definitions are applicable.

Table 5-2:Technical Definitions

DEFINITION	DESCRIPTION
Acid Gas	A gas stream containing predominantly H_2S and/or CO_2 from an Acid Gas Removal Unit
Condensate	A hydrocarbon liquid stream containing Pentane+ compounds
Dehydrated Gas	A gas stream that has been dehydrated to remove water
Dry Gas	A gas stream containing no liquids at flowing conditions, but which may still contain water and hydrocarbons that will fall out if there is a change in flowing conditions.
Feed Gas - Associated Gas	A Gas stream that is produced from an oil well during the extraction of crude oil.
Feed Gas – Non-associated Gas	Hydrocarbon gas originating from gas reservoirs.
GMP 2023	Gas Master Plan 2023

DEFINITION	DESCRIPTION
Lean Gas	Lean nature of gas is defined based C2+ in the gas stream. If C2+ is less than 15% then stream is defined as Lean gas
Mixed Feed Gas Stream	Mixed associated and non-associated gas stream
NGL	A liquid stream with primarily C2, C3, C4 and some C5+ compounds.
Rich Gas	Rich gas is defined based C2+ in the gas stream. If C2+ is more than 15% it is considered rich gas
Sales Gas	Residue gas sent to the Sales gas network.
Sour Gas	A gas stream containing significant amounts of H_2S and/or CO_2.
Sustainable Profile (GMP 2023)	GMP 2023 High Supply Scenario – unconstrained oil production from GMP Guidelines Document from all associated gas fields
Sweet Gas	A gas stream in which the H_2S and CO_2 are not present in significant amounts.
Technical Profile	AON provided profile for technical case for associated gas.
Technical Optimized case	The ADNOC Gas facilities (Das, Asab-0/3, Buhasa, Ruwais, associated pipelines) will be sized to accommodate technical rates from associated gas fields. Habshan facilities will be sized to accommodate technical rate from all associated gas fields (Bab, Buhasa, NEB, Offshore) but not simultaneously. All inlet facilities / inlet compression shall be designed for technical rate.
Unconstrained Production Profile / High Supply Scenario	This shall be based on Upstream's OPEC Unconstrained Oil production case. This scenario corresponds to sustainable projects profiles

5.3 ABBREVIATIONS

ADNOC	Abu Dhabi National Oil Company
AGP	ADNOC Gas Processing
AGES	ADNOC Group Engineering Specifications
AON	ADNOC Onshore
ASME	American Society of Mechanical Engineers
EPCM	Engineering Procurement & Construction Management
FEED	Front End Engineering Design
HVAC	Heating Ventilation and Air Conditioning
IFA	Issued For Approval
IFR	Issued For Review
OEM	Original Equipment Manufacturer
PEP	Project Execution Plan
UAE	United Arab Emirates
UPS	Uninterrupted Power Supply

6.0 PRESERVATION ORGANISATION – KEY STAKEHOLDERS

6.1 Organization

- The P 5.0 DBN Integrated Completion Team owns the Preservation Plan
- P 5.0 DBN Project Leadership Team
- P 5.0 DBN Supply Chain Management
- Equipment Vendor and Manufacturers
- Logistics contractors
- Customs inspections and agents responsible for Customs clearance
- Module Fabrication Yards during the unit and skid assembly
- Site Warehousing and Installation teams

6.2 Construction Disciplines

- Static Equipment, Material Selection and Corrosion
- Rotating Equipment
- Instrumentation & Control
- Electrical Power Electrical Heat Tracing
- Civil/Structural/Architectural
- Process Engineering
- Pipeline

6.3 Tenets of Preservation

Preservation is an essential element of the Facility Design and Construction requirements. As such, P 5.0 DBN needs to demonstrate adherence to a system and establish operational discipline in preservation.

The Tenets of Preservation, like the Tenets of Operation, are based on two fundamental principles:

"Do it safely or not at all" "There is always time to do it right."

Tenets of Preservation
• **Always** identify, discuss, and address potential hazards before beginning ANY preservation activity. • **Always** plan for and use appropriate ladders, scaffold, and harnessing for ANY preservation work to be performed above grade. • **Always** use stop work authority when ANY of these tenets is not followed. • **Always** ensure that anyone in the vicinity of your work knows of your presence, the type of work you are doing, the hazards you have identified, and how to respond in an emergency. • **Always** verify that the work specified in the procedure has been correctly completed to the required standard before signing off the check sheet. • **Always** include guidance on the expected magnitude and frequency of likely environmental conditions for which preservation procedures are intended. • **Always** stop work and re-inspect materials whenever the magnitude and frequency of environmental conditions exceed what was intended in the preservation procedures. • **Always** validate that physical assets and materials remain in the state of preservation required in the Preservation Plan. • **Always** verify that all safety devices on preservation equipment function as designed before commencing any preservation activity. • **Always** verify that relief valves are correctly sized, calibrated, certified, installed, and not isolated before beginning preservation activity. • **Always** verify that the preservation workforce uses appropriate PPE for the preservation task at hand, especially when handling nitrogen or chemicals. • **Always** physically witness work activity when the procedures or check sheets require it. • **Always** repeat the work validation if it was not observed the first time. • **Always** ask questions as needed to make sure you understand how to recognize that work is correctly completed to the required standard before you witness or sign off work. • **Always** validate that active preservation control measures are in place, functioning, and not disabled or bypassed. • **Always** create written procedures to define how and when temporary removal of preservation measures may be approved and how and when they must be restored. • **Always** record preservation status in the check sheet and SCD. • **Always** validate all necessary preservation punch list items are complete before beginning further work. If not, wait to start work until they have been. • **Always** validate that isolation blinds, spades, and valves are in place and the positions specified by the procedure before beginning any preservation activity. • **Always** verify that the specific requirements of the Integrity Management Plan set by regulators have been addressed before beginning any preservation activity. • **Always** follow the Permit to Work, LOTO, SIMOPS, and Isolation procedures in place for the current phase of work. • **Always** create written contingency plans and procedures to address deviations during preservation activities • **Always** follow the contingency plans and procedures developed to address deviations • **Always** record missing, incomplete, incorrect preservation configuration, equipment, or functionality in the punch list • **Always** document any variations of preservation equipment from the design drawings and documents using red-line mark-ups (as-builts). Capture the red lines in SCD and record them as a punch list item. • **Always** create written procedures for preservation activities

Tenets of Preservation
• **Always** read and understand the written procedures, step by step sequence, and work guidance before beginning any work. Always ask questions to ensure understanding. • **Always** follow the written procedures, step by step sequence, and work guidance. • **Always** use check sheets to track and document completion of procedures • **Always** record the check sheets in SCD • **Always** obtain approval from the Preservation coordinator before deviating from the procedure. • **Always** follow the P 5.0 DBN management of change procedure before changing the preservation method or equipment from what is included in the approved procedures or drawings.

6.4 Preservation Inspection Intervals

Period Code	Interval Between Preservation Inspection / Maintenance
0W	Initial Inspection
1W	Weekly
2W	Bi-Weekly
4W	Monthly
13W	Quarterly
26W	Semi-Annual
39W	Quarterly
52W	Annual

6.5 Storage Classification

Classification	Title	Description
1	Outdoor Storage, Non-Covered	Outdoor - When necessary, items shall be stored on pallets or cribbing.
2	Outdoor Storage, Covered	Outdoor - Stored and covered with tarpaulin or shrink wrapping on pallets or cribbing. Allowance for adequate ventilation is provided.
3	Outdoor Storage, Covered, Heat Applied	Outdoors - Stored and covered on pallets or cribbing with a heat source applied to the item to prevent condensation or corrosion. Allowance for adequate ventilation is provided.
4	Indoor Storage	Indoor storage non-climate controlled

5	Indoor Storage, Climate Controlled	Indoor storage with controls installed to maintain temperature and humidity.

7.0 SITE AND LOCATION OF WORK

Habshan Plant is the main central hub for onshore gas processing in Abu Dhabi. This overall gas processing complex consists of associated and non- associated sour gas treatment, condensate stabilization, NGL recovery, sulphur recovery and gas injection units. The overall complex consists of multiple processing trains that have been constructed over the years in line with the growing demand for non-associated gas in Abu Dhabi and the need to process associated gas to enable oil production. Hence, these facilities at Habshan complex are operationally segregated into areas namely Habshan 0, Habshan 1, Habshan 2, and Habshan 3 and Habshan 4

8.0 PRESERVATION ORGANIZATION ROLES / RESPONSIBILITIES

Preservation execution on the Project occurs throughout the project life cycle. Each phase requires a unique focus, and the organization necessary to support preservation will evolve as the project progresses. The contractor/sub-contractor bears the overall responsibility to ensure compliance to the requirements of this procedure. CCC of tagged/maintainable equipment requires the responsible party to provide dedicated preservation personnel, tools, and support equipment to support preservation requirements, oils, grease, and lubricants specified in this plan, scaffolding and access to preserve and inspect equipment, weather mitigation protocols, Co-Console completion of P-ITR's per specified frequencies, and documentation and remediation of issues identified during inspections. This Preservation Plan outlines the organization roles and responsibilities for the Project phases as follows:

8.1 CSU Manager (Required for all periods)

a. Assures Preservation Plan is produced and executed to the appropriate extent for the project.

b. Participates in the preservation planning meeting as required.

c. Obtains approval from the Project Leadership Team on significant deviations from this Project Plan

d. Assures all contracts and ITT documents include preservation requirements for the project.

e. Assures that a preservation audit/verification plan exists, audits/verifications occur per project, and that identified issues are resolved in a timely fashion.

f. If a project is put on hold, the CSU Manager has the responsibility of assuring that the storage areas, Auxiliary Support Systems, regular equipment inspections, and preservation activities are maintained.

8.2 Preservation Manager (Required for all Phases)

g. Holds assigned accountability for the preservation process

h. Leads and documents the preservation planning meeting.

i. Develops the preservation requirements for ITT's and contracts.

j. Monitors all steps of this Preservation Plan and escalates problems to the integrated completion/preservation director as necessary.

k. Audits/verifies the Constructor/sub-contractor receiving, storage, and preservation practices against the plan.

l. Develops and manages the project preservation register/database.

m. Assures that preservation is maintained as the equipment is being installed, and if necessary, issues mitigation plan to restore integrity.

n. Assures a thorough transition to the CSU group at Mechanical Completion and Turnover.

o. Participates in the equipment supplier kick-off meetings to assure the Equipment Suppliers understand the requirement to provide the details of the packaging requirements, shipping, and Preservation Plan.

p. Assures competency of Preservation personnel

8.3 Procurement Representative (Required for all phases)

q. Supports the preservation process by assuring preservation requirements are contained in RFQ/ITT and Purchase Order documentation.

r. participates in the preservation planning meeting.

s. Assures that the contractual wording is unambiguous and allows P 5.0 DBN to enforce the objectives of the Preservation Plan.

t. Assures that the contractual wording does not create conflicts with other contract objectives.

8.4 Engineering Discipline Specialist (Required all phases) (Part-Time Role)

This is often the design engineer, requisitioning engineer, or package engineer.

u. Provides equipment knowledge to the preservation process.

v. participates in the preservation planning meeting.

w. Serves as contributing reviewer for all reviews of the storage and preservation plans for respective equipment.

x. Participates in the equipment kickoff meetings to assure the Equipment Suppliers understand the requirement to provide the details of the shipping and the

Preservation Plan.

y. Serves as an expert technical resource for respective equipment when questions arise from the Preservation Lead.

z. Comments on any proposed deviation or step out from the agreed preservation requirements for supplied equipment.

8.5 Logistics/Customs Specialist

aa. Manages the transportation activity required to deliver materials and equipment to the construction site.

bb. Ensures transporters and logistics personnel maintain preservation integrity during transport.

cc. Manages customs clearance activity, ensuring preservation integrity is not compromised for customs inspections or is restored after that.

8.6 Preservation Lead Material Receiving & Storage

dd. Manage the material/equipment receiving process at the site delivery points. All material/ equipment shall be checked to ensure the supplier's initial preservation is intact

ee. Maintain preservation integrity of hermetically sealed packages, with a humidity detector, by visually inspecting the humidity indicator, noting, and recording the value.

ff. Inspect equipment shipped with an inert gas blanket or under positive pressure, note and record pressures.

gg. Audit the supplier documentation and confirm all WHMIS information has been provided. If deficiencies are apparent, the equipment shall be placed in quarantine until the appropriate information is available.

hh. Manage the handling/offloading activity such that there is no damage to equipment during lifting and on-site transportation.

ii. Attach Preservation labels and tags as required.

jj. Ensure the material/equipment is stored in the proper storage area as per the Preservation Plan.

kk. Report any anomalies to the Preservation Manager for clarification and for necessary actions to be taken for the reinstatement of preservation.

ll. Follow the supplier's instructions noted on the packing crates, and/or in the Dossier. These instructions shall be regarded as mandatory during handling and storage.

8.7 Preservation Planner/Scheduler (Site)

Assures the construction team is prepared to support the Preservation Plan.

mm. participates in the preservation planning meeting.

nn. Assures that materials management, warehousing, and construction staff are aware of their roles and aligned to support the Preservation Plan.

oo. Monitors the storage and preservation requirements of the equipment to ensure that the Constructor's plans are still adequate. Escalates to the Preservation Lead Preservation non-conformances that cannot be resolved.

pp. Assures that the construction processes sustain preservation steps for all equipment and material from receipt of material at the construction site through Mechanical Completion and Turnover.

qq. Communicates the preservation requirements to the construction and fabrication teams.

rr. Monitors the execution of the Compressor Station “Build it Clean” and Clean Construction Program and makes appropriate interventions based on field observations

8.8 Preservation Work Teams

ss. Receives daily preservation instruction from Planner /Scheduler/manager.

tt. Executes preservation activity in accordance with pre-approved preservation procedures and P-ITR’s.

uu. Identifies and communicates preservation nonconformance and rectifies per approved procedures as required.

vv. Updates preservation records and communicate with the Preservation planner/manager when tasks are completed.

8.9 Operations Representative

ww. Assures a long-term focus on preservation and build systems to ensure the operating team is prepared to accept the assets and sustain equipment integrity.

xx. participates in the preservation planning meeting.

yy. Participates in most preservation activities to ensure a long-term focus.

zz. Assists in auditing/verifying the Constructor storage and preservation activities.

aaa. Assures us that the Asset Owner is prepared to sustain equipment integrity after Unit start-up.

bbb. Ensures preservation of capital spare parts is maintained.

ccc. Produces and executes equipment integrity plan after start-up.

9.0 EQUIPMENT SPECIFICATION / FABRICATION PRESERVATION PHASE

9.1 Preservation and Maintenance Setup

a. The preservation manager is assigned early in the Engineering / Procurement phase of the project to ensure project preservation requirements are communicated to all stakeholders.

b. The preservation manger will coordinate with other stakeholders to load “maintainable” tags (including spares) in Co-Console and assign a preservation checklist with inspection frequencies. The preservation checklists will be assigned by function code to individual tagged equipment. Parent tags for packaged equipment, skids, buildings, etc. will be used as the “maintainable” tag with appropriate checklist assigned. Child tags in these instances will be preserved under the main parent tag.

c. Special attention will be given to asset critical packaged and rotating equipment. Vendor special requirements, lubricants and warranty requirements will be considered when developing these preservation checklists.

d. Tagged Assets will be divided into “Critical” or “Non-Critical”. Holidays, weather events, etc. can reduce weekly tag inspections. Critical tags will be prioritized for inspection in these instances.

9.2 Equipment Purchase Order (PO)

a. The home office Procurement / Engineering team will establish preservation requirements. The material requisition documents for equipment and materials are developed by the Responsible Engineer (RE). The RE shall request the storage, preservation, packing and shipping requirements that ensure equipment and material preservation and protection from the Supplier. Technical requirements for preservation are included throughout the PO as part of the Commercial Notes, Technical Notes, specifications, drawings, submittal requirements, etc., The Material Requisitions and resulting Purchase Orders dictate the instructions and submittal requirements for information from the Supplier. Suppliers, including their sub-suppliers will be required to submit instructions for handling, cleaning, preservation (including routine maintenance), packaging, and storage to CONTRACTOR. The key documents submitted by the supplier for review include:

- Inspection and Test Plan
- Manufacturer's recommendation for storage and preservation requirements for equipment and materials, including spare parts.
- Manufacturer's Installation, Operating and Maintenance Instructions
- Packing and Protection Procedure, and any necessary data for transport, handling, and special requirements.

For equipment that is inspected, the Procurement team coordinates with the QA/QC team to ensure Inspection Plans issued to shop Inspectors include requirements to verify preservation prior to shipment. The preservation manger ensures that preservation lubricants/ preservatives along with the required storage duration and type are included in

the PO documentation.

The equipment manufacturer should review Project preservation requirements and submit their comments post-award along with a packing, storage, and preservation procedure (which shall be based on the specified storage duration and type). Preservation requirements are negotiated with the Supplier to ensure adequacy and consistency. Final submission and acceptance of preservation procedures is required prior to shipment of equipment. Suppliers shall include recommended preservation methods during construction.

b. For other EP Contractors, approved preservation procedures with equivalent information must be provided to the COMPANY for supplied equipment prior to shipment of equipment/modules.

c. The Preservation manager reviews the accepted preservation procedures and marked-up standard Preservation requirements submitted by the equipment Suppliers with the assistance of the package engineer.

d. Preservation begins in Supplier facilities prior to shipment. Each Contractor's Inspector/Supplier Quality Representatives (SQR) will monitor and verify a Supplier's initial packaging and preservation activities in accordance with the purchase order requirements prior to shipment.

e. Contractors that are engaged by the COMPANY shall maintain all equipment, materials, components, skids, modules, and spares under their care, custody, and control until turnover to CONTRACTOR. Supplied items shall be turned over to CONTRACTOR with up-to-date preservation and packaging. CONTRACTOR will take over preservation and storage in a "maintenance mode". EP firms shall provide complete instructions for handling, cleaning, preservation (including routine maintenance), packaging and storage from EP Contractors, EP supplier and sub-supplier, procedures, and other related documents and instructions to CONTRACTOR in advance of shipment unless otherwise specified.

9.3 CLEANING EQUIPMENT BEFORE SHIPPING

Immediately upon completion of Supplier quality assurance/quality control (QA/QC) tests and pre-preservation inspections (e.g., factory acceptance tests), equipment and piping shall be thoroughly drained, dried, and cleaned of scale, dirt, and other foreign material.

Liquids used for cleaning or hydrostatic testing shall contain a non-fouling, rust-preventing fluid, or the equipment shall be thoroughly flushed with a non-fouling, rust-preventing fluid.

Suppliers shall ensure that protective coatings used during fabrication such as masking tape, masking paper, etc., are removed from equipment, including areas such as instrumentation, packing gland nuts, followers, seal plates and nuts on mechanical seals.

Equipment meet a standard of cleanliness that is appropriate to the type and duty of the material and shows no visible remnants of protective coatings.

9.3.1 INITIAL PRESERVATION

a. Initial preservation methods shall be developed considering the following:

Shipment mode.

Storage conditions.

Required preservation period.

b. Supplier shall be solely responsible for the following:

Ensuring that preparations are made to prevent equipment deterioration during shipping and storage.

Identifying and making provisions for preservation requirements during shipping and storage, such as shaft rotation, etc.

c. As a minimum, preservation shall comply with the applicable shipping/preservation plan as outlined in the purchase order.

d. The supplier shall be responsible for meeting any preservation requirements in addition to the PSS, as determined by the Purchaser.

e. Only allowed preservatives or a Purchaser-accepted equivalent shall be used. Refer to a list of acceptable preservatives included in this procedure

f. Equipment shall be cleaned and dried before applying preservative materials.

g. Hydrotest water, if present, shall be removed before preservation. After, hydrotesting equipment shall be thoroughly drained and dried. Internals shall be cleaned and preserved according to the approved procedures. Any rags used for cleaning shall be lint-free. Suppliers shall take special care to ensure that inaccessible areas inside the equipment are thoroughly dried, cleaned and preserved in accordance with approved procedures and/or acceptable code/standards. For example: Shell and Tube Heat exchangers shall be drained and dried per Cl. A.8, Annex A, API 660 (9th Edition).

h. VpCI-414 or equal shall be used for cleaning finished, unpainted, exposed metal surfaces before applying required rust preventatives.

i. Measures shall be taken to protect against breakage, dampness, pilfering, and tampering during transportation, handling, and storage.

j. Unless otherwise specified, equipment shall be packaged and weather-protected for transit and for twelve months of outdoor storage in a high-humidity, saline environment.

   Equipment shipped internationally shall have a minimum protection of 18 months of outside storage in a high-humidity, saline environment.

k. If the Supplier cannot practically prepare equipment for such outdoor storage, the Supplier shall advise Purchaser, in writing, of the required special field protection (e.g., indoor storage, outdoor storage under partial shelter, weatherproof outdoor storage, storage under tarps, etc.).

   Equipment shall be referenced to its primary tag (parent), as well as secondary/ancillary

equipment (child) installed on or associated with the main equipment item (e.g., electric motors, batteries, and chargers).

l. Equipment shall be adequately preserved against the entry of dirt or water during shipping and storage.

m. Equipment having stainless steel components shall be protected against contact with chlorides.

n. The supplier shall be solely responsible for the preservation of the equipment until delivery, and for delivering the equipment with initial preservation according to this standard. Any additional requirements shall be submitted, in writing, to Purchaser for acceptance before shipment.

o. Equipment and material shall be handled with care to prevent damage.

p. Equipment cavities, cooling passages, mechanical seals, and positive displacement pump plunger cavities shall be drained of water.

q. Threaded openings shall have plugs of metallurgy equal to the component being capped or plugged. Plastic plugs are acceptable only for nonhydraulic/pneumatic systems. Flanged openings shall be provided with blind flanges, bolts, and gaskets. All temporary shipping covers shall be painted in high visibility orange to indicate removal once equipment is installed. Equipment openings and piping beveled for welding shall be provided with closures designed to prevent contamination with foreign materials and damage to the bevel.

r. When specified in the purchase order, desiccant-bag type protection shall be provided. Desiccant bags shall be securely fastened to the inside surface of the equipment to provide uniform distribution. Quantity of desiccant shall be determined by the Supplier based on surface area, and volume of enclosed space. Equipment shipped with desiccant bags shall be tagged with appropriate red signs identifying that “DESICCANT BAGS ARE ENCLOSED”.

s. Desiccant shall not be in direct contact with metallic surfaces

t. As applicable, Supplier’s short-and long-term preservation procedure shall include detailed requirements for applying and maintaining nitrogen purge, and precautions to be taken. A pressurizing valve and gauge shall be provided and located at an accessible location. A vent valve shall be provided to facilitate venting. Warning label shall be provided identifying as a “WARNING – EQUIPMENT PRESSURIZED WITH DRY NITROGEN”.

u. Unless stated differently in requirements for specific equipment, the following shall apply:

Oil- lubricated bearing housings, seal housings, stuffing boxes, hydraulic equipment, and gear cases shall be fogged and filled to half site-glass capacity with manufacturer-approved oil. Openings shall be closed and sealed tightly.

Externally exposed, bare carbon steel or cast-iron surfaces, including shafts and couplings (except elastomeric components), shall be coated with preservatives, as appropriate. (See Section 15 for details on preservative types.)

Machined surfaces shall be coated with preservatives outlined in section 19.

Grease-lubricated bearings shall meet manufacturer specifications.

v. The supplier shall be solely responsible for the preservation of the equipment until delivery, and for delivering the equipment with initial preservation according to this standard. Any additional requirements shall be submitted, in writing, to Purchaser for acceptance before shipment.

w. Equipment and material shall be handled with care to prevent damage.

x. The supplier shall ensure that preservative and storage lubricants are suitable for the specific application. Examples include the following:

Grease or oil-based products shall not be in contact with components to be installed in oxygen or chlorine service.

Preservatives shall not contaminate interiors of fluor chlorohydrocarbon refrigeration compressors.

Hydrocarbon flush oil shall not contaminate synthetic oil passages.

Note: Preservatives and storage lubricants can adversely affect the safety and operating life of equipment if they react with the Process fluid or operating lubricant.

10.0 PROCUREMENT PRESERVATION PHASE

10.1 Material Receiving, Inspections, and Control-Warehouse and Laydown

a. The engineering, integrated preservation team is responsible for determining the type of equipment that requires a mandatory procurement inspection upon receipt to the project and documents completed inspections in HUB2. Procurement will have inspections of non-maintainable equipment, and those inspections are not required to have integrated preservation team involvement.

b. Upon receipt of a tagged piece of equipment, the procurement team performs commercial receipt inspections, checking the shipment against the packing list and purchase order for count, condition, identification and required documents. In the event the equipment requiring a receipt inspection is packaged for long term storage (e.g., Hermetically sealed package) and does not contain a view window, the procurement team with the assistance of the preservation team will determine if the equipment's receipt inspection can be deferred until the time of withdrawal from the warehouse. If the equipment requires inspection and the sealed package needs to be opened, the procurement team with the assistance of the preservation team shall determine if access doors, shrink

wrapped windows, are required for ongoing preservation activities. Storage conditions of tagged equipment while in the care, custody and control of the procurement team shall follow section 7.3 of this procedure and storage condition documentation provided in the purchase order. As a note, the procurement team shall ensure tagged assets with category 3 storage requirements are connected to power. Light plants, generators, pig tails, fuel, bang boxes, etc. are some of the items that may be necessary for power connections. Regular fueling of assets will need to be considered/planned to include weekends and holidays.

c. Tagged equipment stored outdoors shall not be set directly on the ground. Dunnage will be provided no less than 4” for tagged asset placement.

d. Shrink wrap will cover the top of tagged wooden crates when stored outdoors.

e. Project quality representatives shall develop project specific receiving checklists based on project specific procedures, project specifications and contractual requirements that are included in the HUB2 procurement receiving inspection.

f. The procurement receipt inspection will log delivery date and equipment location to enable the preservation team to initiate the initial preservation inspection in Co-Console.

g. The procurement team shall ensure the receipt inspection is completed within 5 calendar days. Material (especially tagged equipment, spares, special tools) discovered to be nonconforming after inspection shall be controlled and documented on a UOS&D report. Technical assistance for UOS&D disposition shall be provided by the responsible engineer, procurement team, and preservation manager. A non-conformance report shall be generated for conditional release, repair, or use as is. The procurement team will log all defective/damaged equipment on a vendor “back charge” list, inform the vendor of the discrepancy, and withhold final milestone payment until rectification has been completed.

h. OEM equipment shall be maintained and preserved by the sub-contractor responsible until CCC has been transferred to the procurement team.

10.2 Purchaser Responsibilities

i. Purchaser shall be solely responsible for the care and cleanliness of the equipment during offloading, handling, and storage.

- The standard of care and cleanliness shall be appropriate to the type and duty of the equipment.
- Indoor storage shall be used whenever possible
- Outdoor storage is only acceptable for equipment prepared for it by the Supplier.

j. Preservation maintenance shall be performed during the storage period in accordance with this standard

k. Upon receipt, the Purchaser shall immediately inspect equipment and check the

preservation and completeness of the preservation record. See Section 12.1 for more detail.

l. Purchaser shall provide preservation, storage, and preventative maintenance for equipment in accordance with this specification and the equipment specific PSS.

m. Equipment shall continue to be protected and preserved to the fullest extent possible after installation and prior to pre-commissioning.

n. Purchaser shall be responsible for the following (refer to Section 9.0):
   - Preparation of preservation record where applicable.
   - Maintaining preservation records from Supplier.
   - Recording preservation activities on preservation records
   - Updating the computer-based preservation system.
   - Filing and maintaining MSDS in the preservation record for all preservation products used.
   - Continuing preservation maintenance, according to Supplier-provided PSS.
   - Any newly revised requirements due to installation activities.

o. Purchaser shall maintain and update the preservation records as per the requirements outlined in this specification.

10.3 General Requirements

p. Equipment shall be stored according to the requirements specified.

q. Sensitive equipment, including critical electrical equipment, instrumentation, and telecommunications equipment (such as computer systems, control panels, and analyzers), shall be stored in an environmentally controlled building to protect against moisture and condensation.

r. Vapor seals shall be protected until the equipment is serviced for the startup. Damage to the seals prior to startup shall be repaired immediately.

s. Equipment and instrumentation shall be protected from damage due to welding or sandblasting throughout storage and installation periods. As a minimum, a flame-retardant cover shall be provided for protection from welding.

t. Stacking and spacing of equipment and material shall be done in a manner to avoid damage and promote ease of periodical inspection.

u. Temporary protective coverings shall allow free air circulation to prevent humidity condensation and collection of water.

v. Items with machined surfaces shall be stored so that the machined surfaces can be examined periodically for signs of deterioration.

w. Unless otherwise specified, special purpose equipment shall be stored with positive pressure, dry nitrogen purge. A caution sign board shall be attached to the equipment or piping spools to indicate presence of inert gas.
   - Equipment shall have a temporary gauge to determine purge pressure.

- The temporary gauge shall be removed before startup.
- During a periodic inspection, equipment shall be inspected to ensure that purge integrity is maintained.
- If a positive pressure cannot be maintained, purge at a rate of 4–6 standard cubic feet per hour ([SCFH] 2–3 standard liters per minute [SLPM]). Good housekeeping shall be practiced in order to maintain equipment preservation.

x. Mechanical equipment may be stored outdoors with appropriate preservation

y. Equipment and material shall be stored free from direct ground contact and away from areas subject to ponding water.

z. As a minimum, laydown areas shall be covered with gravel.

aa. For outdoor storage, cross-cut timber with at least a 4-inch × 4-inch (89-mm × 89- mm) nominal cross-section, laid flat and level, shall be used for laydown.

- Equipment weight shall be considered when selecting timber size.
- Warped timber or telephone poles shall not be acceptable.
- Timbers shall be placed perpendicular to major support structures and shall be the full width of the skid or baseplate.

bb. In addition to the preservation requirements, plastic tarps may be used to cover equipment in outdoor storage. Plastic tarps shall not be used in lieu of preservation requirements.

cc. When it is necessary, it shall be covered by canvas or tarpaulin at laydown area until installation and preserved in accordance with instruction by its respective supplier. All vegetation will be removed from the location and kept clean and free at all times, thorough inspection regimes will be developed, and results logged.

10.4 Weather Sensitive Materials

Weather sensitive materials, equipment and supplies shall be stored inside containers or other weatherproof structures. Items such as instruments, small tools, delicate equipment, and specialty items must be weather protected and stored indoors under special conditions as laid out within the supplier specifications and guidelines.

10.5 Gas, Electrical, and Special Services Required

One or more of the following types of service, if required, shall be provided for the preservation and storage of any particular equipment item:

a. Nitrogen gas service for equipment or materials requiring a nitrogen gas purge to be maintained (e.g., heat exchangers).

b. Electrical service for equipment or materials requiring electrical power to energize space heaters.

c. Unique service for equipment or materials having special requirements as defined by Supplier recommendations.

10.6 Requirements for Alloy Material

a. A segregated and secured receiving and storage area shall be created and maintained for alloy bulk materials.

b. Upon receipt of alloy material, if required, warehouse personnel shall inform the responsible QA/QC inspector for positive material identification (PMI).

c. Material shall be staged for inspection until it is released by the QA/QC inspector for storage in a secure and segregated location.

d. Alloy materials shall be maintained in secure and segregated locations, as reviewed, and accepted by QA/QC, until issued to the installing contractor.

e. Mark numbers for shop-fabricated pipe spool pieces shall be verified upon receipt.

- A grid system for the spool laydown yard shall be developed for location purposes.
- The location of each spool shall be noted on the packing list for input into Purchaser's materials management system.

f. Verification that each valve and specialty item comply with tagging requirements, as detailed in the purchase order (PO), shall be performed for each item received.

g. Carbon and low alloy steel shall be protected from any contact with corrosive atmospheres or wet atmospheres so as to prevent rust formation.

10.7 Material Receiving, Inspections, and Control-Site

The same general rules apply to equipment delivered directly to site as outlined in section 12.1 of this procedure including storage condition requirements. The procurement team is expected to conduct a receiving inspection for tagged items delivered directly to site where CCC has been transferred to their purview. Construction/sub-contractors shall be prepared to have power supplied for all category 3 items (bang boxes, light plants, generators) and scaffolding teams available for preservation support(access). Tagged items with category 5 storage requirements shall be sent to site for immediate installation. Tagged items sent to site with category 4 storage requirements may need to be covered with shrink wrap, placed in a connex, etc. before installation. See section 12.8 below for reference.

10.8 Storage and Maintenance of Equipment-Warehouse and Laydown

a. The storage requirements for equipment shall be followed in accordance with standard Field Material Storage Control or Supplier's recommendations, whichever is more stringent.

b. Typical maintenance inspection requirements include:

   i. Proper storage condition on a flat level, well drained, well compacted surface 4-6" off the ground (Monthly-4W)

ii. Provided with weather protection with minimum 6" air space between equipment (Monthly-4W).

iii. Visual inspection for physical damage, corrosion, or deterioration (Monthly-4W).

iv. Cleanliness is maintained (Monthly inspections-4W)

v. Identification tags/marks are in place and are securely fastened (Initial Inspection-0W)

vi. Protective seals and covers are intact (Initial inspection-0W)

vii. Protective coatings, lubricants, and preservatives are intact (Initial Inspection-0W)

viii. Periodic replacement of oils or preservatives as required by Supplier (18-24 month-180/365)

ix. Humidity indicators are monitored, and desiccant is changed as required (Monthly Inspection-4W).

x. Inert gas blanketing is maintained within specified limits. Any leaks should be immediately corrected. If leaks cannot be corrected, the appropriate Supplier should be contacted Monthly N2 Inspections-4W)

xi. Space heaters are energized. Power will be required (Weekly inspection-01W)

xii. Electrical insulation resistance tests are being performed as required (Annually-52W)

xiii. Rotation of shafts on rotating equipment is occurring, as required (Monthly-04W).

10.9 Preservation and Maintenance Execution-Warehouse and Laydown

NOTE: Job hazard analysis review of preservation activities will be submitted for EHS approval with safety mitigations identified. Preservation ToolBox talks and Preservation 411's will be conducted prior to the beginning of each shift and referenced on Project STARRT cards.

a. The procurement team via HUB2 shall ensure delivery date and locations are correct. RFID tags/barcode stickers along with "laydown yard grids" shall be used to ensure location accuracy.

b. Tagged equipment storage locations will be reported as acceptable/unacceptable during preservation inspections.

c. Co-Console shall list upcoming tagged inspection activities in 14-day increments.

d. Completion of the preservation and maintenance activity signs off on the preservation inspection checklist shall be completed via the preservation app. The preservation individual that completes the task will add notations and pictures on the PM inspections along with any identified defects or issues.

e. The preservation manager will ensure protocols are setup up to flag and track any inspection that has issues identified during the preservation inspection activity.
f. Subsequent work orders will be generated to remedy issues found during preservation inspections.
g. Preservation dashboards will be created for preservation progress tracking.
h. Audit processes shall be created via power apps, etc. to ensure preservation protocols and processes are being followed.
i. Preservation of tagged/maintainable equipment under the CCC of procurement is the responsibility of the integrated preservation team.

11.0 TRANSPORTATION TO CONSTRUCTION AREA PRESERVATION PHASE

Tagged/maintainable items loaded for transport to the construction work front shall not have preservation removed. The procurement team shall update location status as “Issued” in SiteSense. The construction area/sub-contractor requesting equipment that requires preservation shall be ready to install the equipment in its permanent plant location within 7 days of equipment withdrawal. Transportation vehicles, construction gates, lunch tents, etc. shall have SiteSense tracking arrays installed to ensure tagged items with RFID tags can be located upon arrival at the construction work front.

11.1 Pallets

Small, tagged items will be placed on wooden pallets for shipment to the construction work front. Small, tagged items shall not be placed directly on truck beds without the use of pallets, softeners, etc.

11.2 Rigging

Lifting lugs, etc. shall be used for equipment loading/unloading. Rigging directly to valve actuators shall not be permitted for valve loading/unloading.

12.0 CONSTRUCTION AREA PRESERVATION PHASE

12.1 Scaffolding

Construction/sub-contractors shall provide scaffolding access as necessary for preservation inspection activities. Lifts may be used in lieu of scaffolding with safety approval. Scaffolding access required for preservation shall be signed off for both dayshift/nightshift and weekends as necessary.

12.2 Power Requirements

Construction/sub-contractors shall provide power supply/connections to tagged equipment requiring storage condition three requirements. Construction/sub-contractors shall make available

light plants, generators, bang boxes, pig tails, etc. for storage condition three requirements. Equipment used for providing temporary power shall be rated for outdoor use. Fueling of temporary equipment and maintenance of temporary equipment used for preservation is the responsibility of the indirects team. Fueling/maintenance of equipment used to provide temporary power shall be supported on both dayshift/nightshift and weekends. Construction/sub-contractors shall have spare equipment for supplying temporary power on hand to replace equipment with maintenance issues. Equipment used to provide temporary power shall have signage affixed to the equipment indicating “preservation use only” along with a current contact name and number in the event issues are found with the equipment. Cables and pigtails used to provide temporary power will also have a “preservation only” tag affixed to each cable. Cables shall be routed using cable trees and S hooks to ensure cabling is kept off grade and properly elevated to ensure pedestrian access is not limited or restricted.

12.3 GAS (Grout/Alignment/Support)

Before piping to tagged equipment, construction/sub-contractors shall follow GAS (grout/alignment/support) protocols. Grouting must be completed, initial alignment must be completed, and first fixed support must be installed BEFORE piping to equipment. Protection of connections during alignment shall follow section 19.1.

12.4 Release of Nitrogen Purge from Equipment

Construction/sub-contractors shall inform the integrated preservation team of intent to release N2 purge from equipment for piping purposes. **The following guidelines shall be followed when removing nitrogen purge from equipment:** Establish 10-foot exclusion zone around flange opening to be vented. Vented flange is preferred to be at highest point at gauge attachment.

Apply snoop to flange, loosen bolts of flange, listen for N2 escaping. Do not directly breath escaping nitrogen (SAFETY precaution – Make this clear that only competent persons in well-ventilated area shall perform de-pressurization). Continue to snoop until bubbles do not appear while monitoring gauge until reading is ‘0”. Check for N2 presence using N2 gas detector/sniffer.

Remove N2 purge caution tags from openings and remove exclusion zone red barricade tape.

Apply N2 removed tags to all openings.

Preserve flange openings per project flange management plan. Flanges should comply with B16.5/.47/PCC-1 acceptance criteria.

If equipment is re-charged with nitrogen, the following guidelines shall be followed: Establish a 10-foot exclusion zone per 4.1 under precautions definitions. Ensure all pathways are isolated/properly closed before commencing N2 charging activities.

Install regulator on Nitrogen source. Regulator control should indicate “0” when the bottle is turned on.

Connect N2 Charge/Gauge Kit to the Nitrogen bottle regulator.

Connect pressure hose to N2 charge/gauge kit and N2 bottle regulator.

Open valve on N2 charge/gauge kit

Open valve on N2 bottle and increase N2 regulator pressure to approximately 5-14 PSI / 0.35-1.0 BAR to bleed air from hose assembly – allow N2 to bleed for 3-5 seconds – close N2 Charge/Gauge kit bleed valve. The charging hose is now air free.

Close the valve on the equipment to be charged (1st) and then the N2 bottle valve (2nd) – hold for approximately 1 minute – while charging hose and connections are holding, check for leaks between Nitrogen supply and vessel. Utilize “Snoop” or an alternative soap/leak testing product that produces bubbles. If any leaks are present, correct before proceeding – If no leaks are present, proceed with N2 recharging of equipment.

If no leaks are present in connections between Nitrogen supply and equipment, open valve on Nitrogen supply and valve on skid/equipment to be recharged (regulator should be set to 5-14 PSI / 0.35-1.0 BAR). Slowly open valve on equipment and allow equipment to completely fill/recharge at 5-14 PSI / 0.35-1.0 BAR. When pressure stabilizes, check all flanges/openings for leaks with “Snoop” or an alternative soap/leak detection product – tighten / stop leaks as needed before proceeding to fully recharge equipment.

If recharge connection is not at the highest point, a high point bleed point must be utilized to fully exchange air for nitrogen while filling equipment internal space with nitrogen.

If skid/equipment requires removal of cover to replace gasket to reseal, the skid/equipment will require the fill/bleed operation to be performed for 3 cycles, before recharging for preservation (this is to verify removal of all moisture laden air).

When N2 equipment purge has been recharged to 80-90% of max volume, recheck for leaks using SNOOP product or equal. If no leaks are found, recharge fully – if leaks are present repeat snoop/tightening as all leaks must be stopped before proceeding.

Recharging cycle of filling and bleeding should be performed at least 3 cycles. If skid/equipment has been left open for more than 24 hours and/or open during inclement weather – during first 3 cycles, bleed off nitrogen at low point and monitor for liquid drainage. If liquid is present, re-dry interior prior to re-purging activities.

Ensure N2 purge caution tags are affixed to covered openings and remove barricade tape.

Monitor and record N2 pressure readings weekly per Co-Console ITR.

IF N2 cannot restored, alternative N2 purging solutions may become necessary. See section 19.1 for reference

12.5 Interconnection of Module-to-Module Piping

Flanged and beveled protection will be removed for module-to-module welding/flange connections. Construction/sub-contractors shall wrap pipe ends and flanges until final connections are completed.

12.6 Preservation and Maintenance Execution-Site (Construction Areas)

Preservation protocols outlined in sections 12.8/12.9 shall apply. The main risk to tagged equipment during the construction phase is water/foreign matter inclusion due to open equipment and connections.

12.6.1 JHA's/ToolBox Talks/Preservation 411

Job hazard analysis review of preservation activities will be submitted for EHS approval with safety mitigations identified. Preservation ToolBox talks and Preservation 411's will be conducted prior to the beginning of each shift and referenced on Project STARRT cards.

12.6.2 Preservation Areas

Preservation teams in each construction/sub-contractor area will have dedicated preservation locations for consumables, tool, equipment storage, tents, trailers, etc.

12.6.3 Preservation during lunch

Equipment, spools, etc. actively being worked by construction/sub-contractors shall be covered during the specified lunch shift. See section 19.1 for a list of acceptable preservation items

12.6.4 Weather Mitigation

Before project release for weather events, construction/sub-contractor areas will ensure equipment, spools, etc. are protected per section 19.1.

12.6.5 Preservation Craft

Craft solely dedicated to preservation will be provided by each construction/sub-contractor area. Examples of craft include…millwrights, electricians, fitters, laborers, and indirect labor for preservation support (carpenters/teamsters, etc.). Dedicated preservation craft and all preservation related consumables, equipment, scaffolding, etc. shall be supplied by sub-contractors where CCC is specifically stated required per contract. Current preservation contact lists shall be provided for each area. Dedicated integrated preservation craft shall be assigned to BOP construction areas and supported with general foremen and have a dedicated preservation superintendent or equal.

12.6.6 First Fills

Preservation oils shall be removed before handover to Pre-commissioning. Project approved "operational" oils and greases will be installed, documented, and tagged prior to construction turnover. Used/Spent Preservation oils and greases shall be disposed of/recycled properly per project guidelines.

12.6.7 Preservation Recycling

End caps, pipe COVERS, flange BUMPERS, etc. shall be returned for reuse to the preservation storage area. Preservation recycling bins will be staged at construction elevations/locations for weekly pickup.

13.0 TESTING & RESTORATION PRESERVATION PHASE

13.1 General

Tagged items dropped for testing (control valves, PRV's, etc.) shall be preserved with flange DOTS+VCI, placed on dunnage and covered, or sent back to the warehouse until final restoration (see section 19.1 for reference). PRV's shall be kept upright upon removal. Flange DOTS+VCI/flange BUMPERS must be installed immediately after test blind removal to ensure test package stays clean, dry, and closed.

13.2 CSU Leave Out Lists

CSU may request certain tagged items not be restored depending on CSU cleaning activities, etc. CSU will communicate leave out requirements to the testing team and will take care, custody, and control for tagged items not restored by construction/testing team.

13.3 Purging and Drying

Test packages related to process fluid codes as well as suction sides of rotating equipment shall be confirmed clean and dry by construction (ITR, borescope, witness, etc.) prior final restoration.

14.0 PRE-COMMISSIONING / COMMISSIONING / STARTUP PRESERVATION PHASE

14.1 Pre-commissioning

Preservation activities conducted during the construction phase of preservation shall continue during the Pre-commissioning phase until expressly directed by CSU. The entity responsible during the construction phase shall continue to preserve equipment after MC until ready for start-up or as expressly directed by iCSU. Work Authorizations (WA), Lock out tag out (LOTO) and any other specific training will be required before performing preservation activities. Co-Console shall be used as the active database for preservation activities. Tagged assets associated with system handover will be removed from "construction" designation on the preservation dashboard and referenced as "CSU" for ITR tracking/completion purposes. Cleaning and flushing activities will be conducted in accordance with applicable procedures. Tagged equipment may be left in place or removed for flushing/cleaning activities.

j. Proving and validating the facility and equipment and associated components are:
   i. Functioning
   ii. Ready for Dynamic Commissioning
   iii. As Pre-Commissioning of a system progresses, an interim certificate may be issued in order to allow Commissioning work to commence. This is handled in the form of a

Functional Testing Certificate (FTC) which denotes that all equipment within a particular System has been successfully functionally tested.

iv. When all System Pre-Commissioning has been completed, an MCC/RFC Certificate may be raised to denote that the System is deemed complete and safe to introduce process fluids and media into Systems deemed complete. As start-up cannot commence until all works have been completed for the System, an updated consolidated punch list will be collated and submitted for review with the dossier.

14.2 Commissioning

Commissioning specific ITR's will be loaded in Co-Console for preservation. Preservation ITR's in the commissioning phase of preservation will include specific/dynamic activities that support commissioning AWP and sequencing. Commissioning activities will follow the project approved system handover skyline. Commissioning activities shall ensure equipment and components are functioning as an operational system, systems function per specifications, and systems are ready for start-up and operation. When a System has been Commissioned and is in a state of readiness for the introduction of hydrocarbons, the Ready for Start-up (RFSU) Certificate will be issued. This certificate illustrates that all relevant activities are complete and that any outstanding issues, queries and punch lists are identified for review.

14.3 Startup

k. Start-up & Handover - Introduction of Hydrocarbons and Performance Testing to confirm steady state operations and the facility meeting design and operational specifications

l. Operational Support - Post Start-Up support to ensure final acceptance criteria are met with Handover to Operations.

m. Final Acceptance Handover Certificates confirm that the milestones highlighted have been achieved and are supported by accompanying documentation.

15.0 DOCUMENTATION

15.1 Equipment Suppliers Procedures

All procedures specified by the equipment suppliers shall be considered. The suppliers shall be required to provide procedures to be followed for the protection of materials or equipment for a minimum one-year storage period. Equipment supplier's concurrence shall be obtained for the use of any preservatives or temporary lubricants etc., to maintain the supplier's warranties for the equipment during transportation, erection and preservation of material or equipment.

15.2 Specific Preservation Procedure

Offsite fabricators/sub-contractors shall prepare a procedure for preservation. The specific preservation procedures shall describe the selection of the preservatives, define the frequency of application, requirements for turning of rotating equipment or stroking of valves, and the need to remove preservatives prior to pre-commissioning, mechanical completion and start up. If supplier

specifies preservatives related to maintaining warranty, the recommendation shall be incorporated into the specific preservation procedure.

15.3 Preservation Record and Activity

When preservation activity is necessary for specific material during module fabrication, construction and storage at module fabrication yard/Site, log sheets for respective equipment / material shall be prepared for evidence of the periodical preservation activities. Module fabricator/sub-contractor shall report to CONTRACTOR when any damage to items or missing items that may impact the preservation method has occurred or is found.

n. Preservation records shall be completed by the Supplier and Preservation Team.
o. Each piece of equipment shall have a preservation record that includes the following:
   - Preservation records
   - Tagged equipment inspection steps for each maintainable item
   - Records showing issues on tagged equipment have been rectified
p. Stand-alone tagged equipment shall have one preservation record.
q. Bulk equipment orders (e.g., pipe, valves, untagged instruments, etc.) shall not be required to include a preservation record sheet for each item composing the shipment.

15.4 Preservation Record Sheets

Physical documentation and recording of preservation activities shall be carried out using preservation record sheets

r. The supplier shall generate preservation record labels and affix to appropriate equipment prior to shipping.
s. Preservation labels shall be affixed to equipment as follows:
   - Stand- alone tagged equipment shall have one preservation label for each 316 stainless steel tag.
   - Bulk equipment (e.g., pipe, valves, untagged instruments, etc.) shall not be required to have individual preservation tags.
   - Prior to shipment, the Supplier tagging plan shall be submitted to the Purchaser for acceptance.

15.5 Electronic Recording

t. An electronic-based system shall be used to administer the periodic preservation requirements of the equipment.
u. Purchaser shall perform the following functions:
   - Maintain and update the electronic preservation system.
   - Report the initial preservation status by signing the applicable preservation record sheets.
   - Register the status indicated on the preservation record sheets into the preservation record upon receipt.

16.0 PREFERRED PRESERVATIVES AND USES

The following table lists the preferred preservatives; however, a Purchaser-accepted equivalent may be used if approved by P5.0 DBN Preservation Manager. Preservatives in the table below shall be approved and on file with EHS before use.

Type of Preservative	Equipment	Purpose	Preferred Preservative
Oil Additive	Oiled Equipment Bearings	Used as a vapor corrosion inhibitor. This preservative shall be an oil-based vapor corrosion inhibitor blended with the applicable lube oil in accordance with the manufacturer's recommendations.	**CORTEC M-531(5%) mixed with ISO 32 (95%) oil at a 95-5 ratio.**
Grease	Greased equipment bearings	VpCI infused grease	**CorrLube or Equal**
Long Term Outdoor Protection	Bright metal surfaces (e.g., shafts, couplings, pump feet).	VpCI 368 is a solvent-based, temporary waxlike coating that provides protection in harsh, outdoor, unsheltered applications	**CORTEC VCI-368 or Purchaser-accepted equivalent.**
Flange Faces/Weldable Primer	Corrosion Protection of Flange faces, nozzles, pipe bevels	Outdoor water based rust preventative. Provides a fast drying, water-based temporary coating that provides protection in harsh, outdoor, unsheltered applications. Provides multi-metal protection. Excellent UV resistance	**VpCI-391 or equal**
Rust Remover	Removes rust form ferrous and non-ferrous metals	Used as a rust remover. Prevents flash rusting. VpCI 423 is in gel form.	**VpCI 422/423 or equal**
Passivation	Stainless steel passivation	Clean and passivate stainless steel with citric acid. Citrisurf 77 will not remove rust. Citrisurf 77 Plus will remove rust.	**Citrisurf 77/Citrisurf 77 Plus**

Type of Preservative	Equipment	Purpose	Preferred Preservative
Fuel additive	Diesel Firewater Fuel tanks, etc.	Multifunctional fuel additive. It serves as a corrosion inhibitor, fuel stabilizer, and water emulsifier for gasoline, diesel, and gasohol mixtures.	**VpCI-705 or equal**
N2 Alternatives	VpCI 308-Vessel, Exchanger, Compressor N2 Alternative VpCI-337-Turbine N2 Alternative (hot end)	Self-Healing corrosion inhibitors	**Pouch-VpCI 308** **Fog-VpCI-337**
Cleaners/Degreasers	Cleaner/Degreaser	VpCI-414 coating remover will not affect painted surfaces. VpCI-239 Cleaner and corrosion protection. Can be used directly on all electrical surfaces (motors, wiring, junction boxes, etc.)	**VpCI-414 or VpCI-239**
Shrink Film	126: Spare parts (bearings), motors, insulation transitions, etc. Dr. Shrink: Access doors for crates, small coverings	VpCI-126 Corrosion inhibiting shrink film	**VCI- VpCI-126** **Non-VCI: Dr. Shrink 6-8ml**
Flange Protection/Cleanliness	Flanged ends, nozzles, etc.	Adhesive is embedded with a corrosion inhibitor. ANSI sizing through 36". Above 36" pressure class is required	**flangeDOTS+VCI** **flangeDOTS+VCI+Tag**
Flange/Nozzle/ Valve Impact Protection	Flanged ends, nozzles, etc.	HDPE protectors. Will not warp crack or absorb water.	**Low: flangeCORR** **High: flangeBUMPER** **Critical: flangeBUMPER+Gasket**

Type of Preservative	Equipment	Purpose	Preferred Preservative
VpCI Detection	Detects the Presence of VpCI vapors	Test strip will change from blue to pink/red when VpCI is present	**Cortec VpCI Detection Kit**
Isolation Protection	Flanged ends, nozzles, etc.	Can be used in lieu of a spiral wound gasket for alignment. Isolation of pipe during air blows	**mockGASKETS**
Beveled Pipe	Beveled in spools CS/SS	Closure of beveled end pipe	**HDPE (Under 24") M&P or equal** **Foldable over 24": P3 pipeCOVERS or equal**
Flanged Openings Weather Mitigation	Flanged Opening Protection	Keeps flanged openings clean, dry, and closed during weather events, end of shift, and lunch time. paddleMAGS can be used to protect flanges when rolling out valves	**SS-flangeDOT+VCI+Tag** **CS-paddleMAG**
Emitters	Corrosion protection for metal components and parts enclosed in non-ventilated control boxes, cabinets, panels, etc..	VpCI-101 (1 cubic foot of protection) VpCI-105 (5 cubic feet of protection) VpCI-111 (11 cubic feet of protection) Product / Sizes in Inches (Centimeters) / Protection Volume 131 / 2 x10 (5.08 x 25) / 1.5 ft³ (.042 m³) 132 / 10x10 (25x25) / 8 ft³ (0.23 m³) 133 / 1x1 (2.5x2.5) (adhesive back) / 0.25 ft³ (7L) 136 / 1x 1 (2.5 x 2.5) / 0.25 ft³ (7L) 137 / 130' x 4.5' (39.6m x 1.35m) / 10ft³/ft² (3m³/m²)	**VpCI-101** **VpCI-105** **VpCI-111** **130 Series Squares**
Tape	Beveled Spools, etc.	Affix end cap/pipeCOVER to spool	**pcTAPE or equal**
Desiccant	**Humidity Control in closed spaces**	STRING SEWN CLAY DESICCANTS MODEL NO. / BAG DIMENSIONS / UNIT SIZE / PROTECTS CU. FT. / BAGS/ DRUM / LBS./ DRUM S-11204* / 4 3/4 x 9 1/2 x 1 1/2" / 16 / 13.33 / 150 / 188 S-2854 / 8 3/4 x 12 1/2 x 2 1/4" / 80 / 66.67 / 30 / 194 TYVEK® BAG CLAY DESICCANTS - 34 GALLON DRUMS MODEL NO. / UNIT SIZE / BAG DIMENSIONS / PROTECTS CU. FT. / BAGS/ CONTAINER S-8029 / 1 / 3 x 4 x 1/4" / .83 / 1,300 S-8030 / 2 / 3 x 6 x 3/8" / 1.67 / 800 S-8031 / 4 / 5 x 5 1/2 x 1/2" / 3.33 / 500 S-1605 / 8 / 5 x 8 x 1 1/8" / 6.67 / 300 S-1606 / 16 / 5 3/4 x 10 x 1 1/2" / 13.33 / 150	**Uline or Equal**

Type of Preservative	Equipment	Purpose	Preferred Preservative
		TYVEK® BAG CLAY DESICCANTS - 5 GALLON PAILS MODEL NO. / UNIT SIZE / BAG DIMENSIONS / PROTECTS CU. FT. / BAGS/ CONTAINER S-5163 / 1/6 / 1 x 2 1/2 x 1/4" / .14 / 1,200 S-5164 / 1/3 / 1 x 3 1/2 x 1/4" / .28 / 700 S-5165 / 1/2 / 3 x 3 x 1/4" / .42 / 550 S-5166 / 1 / 3 x 4 x 1/4" / .83 / 300 S-5167 / 2 / 3 x 6 x 3/8" / 1.67 / 150	

17.0 PRESERVATION SEQUENCING

On Hold. Will develop along with GCC once GCC is finalized.

18.0 APPENDIX 1 – EXAMPLE OF SHRINK WRAP PRESERVATION

19.0 EXAMPLE OF PRESERVATION ITR (P-ITR)

E04P		CKDBGR
	Electrical Motors and Generators	

Item / Tag No.	Dummy	Description	Test
Manufacturer		PO No.	
Model No.		Serial No.	
Location			
Drawing Reference / Rev.			

No.	Description of Task	OK	N/A	Act
1	Confirm units are clean and, where appropriate, stored in air conditioned room keeping equipment dry, or as a minimum inside a weather proof dry warehouse. Storage environment should be 15°C minimum and have 40-60% relative humidity.	☐	☐	☐
2	Confirm motors have been insulation tested on receipt, all phases to earth and where possible phase to phase.	☐	☐	☐
3	Confirm insulation is re-tested and recorded (Use E05P) whether the motor is stored in a warehouse or installed on site. Test Voltage < 500V Report if IR reading is< 100 MΩ for LV generators. < 150 MΩ for HV generators.	☐	☐	☐
4	Confirm power supply connection for space heating is glanded and clearly marked.	☐	☐	☐
5	For heaters that are not temporarily energized, measure insulation resistance phase to phase and phase to earth.	☐	☐	☐
6	Confirm that non terminated cables are fitted with shrinking shroud.	☐	☐	☐
7	Check equipment is free from mechanical damage.	☐	☐	☐
8	Inspect connection/junction boxes for moisture and replace corrosion inhibitor if necessary. Ensure flame paths are clean & have approved grease for certified Exd equipment	☐	☐	☐
9	Inspect for corrosion, cleanliness and moisture, add preservative as required and verify sealing and protection.	☐	☐	☐
10	Check that greased bearings contain adequate grease.	☐	☐	☐
11	Check oil bath bearings and housings are preserved and oil drain and inlet are plugged with metal or plastic plugs.	☐	☐	☐
12	Rotating parts move one and 1/4 turn. Ensure that the new resting position varies from check to check.	☐	☐	☐
13	Confirm all cabinets, panels, junction boxes, enclosure doors and covers have not been damaged during the inspection checks. Confirm that they are replaced / installed correctly.	☐	☐	☐
14	Confirm preservation tag or label is fitted & updated.	☐	☐	☐

Comments:

	Completed By	Accepted By
Print Name:		
Company:		
Date:		
Signature:		

20.0 PRESERVATION COORDINATION WORKFLOW

Systems Completion Management System

Preservation Workflow

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OBJECTIVE: To define the overall Systems Completion Strategy, processes, interface and associated Roles & Responsibilities across all Phases of the Project.

	RESPONSIBILITY / INSTRUCTION	Package Engineer	Commissioning Manager	Certification Lead	Discipline Leads	Preservation Lead	Warehouse Supervisor	Preservation CONTRACTOR	Turnover Coordinator	Operations	Notes: 1. The Commissioning Consultant aids in the set up of the Initial Contract, prepares the basic philosophy and then recruits the Team based on Scope Of Work agreement. 2. Many of the Instruction items below may be done by the Client organization and should be reviewed after Project set up and aligned with the more detailed RASCI Matrix
1	Define Device Types that require Preservation Agree level of Preservation tracking										Review Facility Asset tables / Tag Register and define which Tagged Equipment types require Preservation tracking and Validation. NOTE: This is usually coordinated in accordance with RAM, CMMS and SPARES to define Critical devices
2	Review Vendor Preservation Guidelines Obtain Vendor Preservation Guidelines for identified Equipment										Review Vendor Preservation Guidelines and WARRANTY Requirements
3	Create Preservation Routines Review GoPreserve® P-ITR's & GAP to Vendor recommendations to create the suite of P-ITR's for the Facility										GAP Vendor Preservation Guidelines & create Preservation Routines (P-ITR's). P-ITR's identify Preservation activities, Frequencies, Preservation mediums / fluids and Tools
4	Populate GoPreserve(r) Import Equipment tag types into GoPreserve(r)										Populate GoPreserve® and create 7-Day and 14-Day Preservation Lookaheads and Weekly Preservation Execution KPI's.
5	Receive Tagged Equipment on Site Receive Equipment, issue OS&D if applicable and manage Preservation until issue to Site										Issue Daily report to the Preservation Coordinator to log Receipt of Tagged Equipment in GoPreserve® which sets the Preservation Clock / Frequencies
6	Install Equipment & Preserve Request Equipment from the Warehouse, Install and Execute Preservation										Execute Tagged Equipment Preservation in accordance with the P-ITR's and Frequency. Complete P-ITR's for Daily upload of ongoing Preservation activities up to the System TCCC to CSU.
7	Preservation Audit & Validation Field verification of Construction Contractor Preservation										Field Validation of Tagged Equipment Preservation, resolve any Issues with Non-Compliance and resolve Issues related to Preservation tools, materials and Personnel to ensure Preservation Frequency is complied with
8	Commission Equipment & Preserve Continue Preservation Coordination after System TCCC until Handover										Execute Tagged Equipment Preservation in accordance with the P-ITR's and Frequency. Complete P-ITR's for Daily upload of ongoing Preservation activities up to the System TCCC to CSU.
9	Preservation Audit & Validation Field verification of Preservation during Commissioning until Handover										Field Validation of Tagged Equipment Preservation, resolve any Issues with Non-Compliance and resolve Issues related to Preservation tools, materials and Personnel to ensure Preservation Frequency is complied with
10	Preservation Certification Ensure that all Preservation Certification and history is available from OEM Vendor Ex-Works until Handover										Collate all Preservation Certification and history for all Tagged Items within the Sub-System that forms the HANDOVER to Operations. NOTE: Ensure that Warranty Expiry date is defined by Equipment
11	Operate & Maintain Ensure that all ORA activities are in place in advance of System HANDOVER.										Validation that CMMS Routines, SPARES and Operations Training has been complete and in place for System HANDOVER

- Yellow fill / hatching indicates that a record is required,
- No fill indicates that no record is specified

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